Cluster Member Galaxy Angular Density vs. Fiber Density: Implications for Spectroscopic Completeness in Stage-V Surveys

Spec-S5 White Paper — Working Draft

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1 Motivation

Multi-object fiber spectrographs allocate a finite number of fibers over a fixed field of view, setting an effective *fiber density* (targets arcmin^{-2}). When the projected angular density of cluster member galaxies exceeds this fiber density, a single-pass observation cannot obtain spectra for all members simultaneously; multi-pass strategies or fiber-priority schemes become necessary.

This note quantifies the member-galaxy angular density for a well-characterized, low-redshift cluster sample and compares it with the fiber densities of two representative spectroscopic surveys:

- **DESI**: $\sim$5,000 fibers over a 7 deg^2 field of view $\Rightarrow 0.198\,\mathrm{arcmin}^{-2}$;
- **Stage-V (Spec-S5)**: $\sim$20,000 fibers over a 5 deg^2 field of view $\Rightarrow 1.111\,\mathrm{arcmin}^{-2}$.

Understanding where the cluster angular density falls relative to these thresholds informs whether satellite galaxy spectroscopic redshifts can be obtained "for free" during the primary survey or whether dedicated multi-pass observations are required.

2 Data

We use the SDSS DR8 red-sequence Matched-filter Probabilistic Percolation (redMaPPer) cluster catalog (Rykoff et al., 2014):

- **Cluster catalog** (`redmapper_sdss_dr8.fit`): cluster ID, photometric redshift z_λ, and richness λ.
- **Member catalog** (`redmapper_sdss_dr8_mem.fit`): member galaxy ID, parent cluster ID, and projected cluster-centric distance R in units of $h^{-1}\,\mathrm{Mpc}$.

Both catalogs are publicly available through VizieR (J/ApJ/785/104, Tables 1 and 2). We restrict the analysis to clusters at $z_\lambda \leq 0.5$, yielding 23,801 clusters.

3 Method

3.1 Angular density definition

For each cluster we compute a projected angular density Σ_θ as:

$$\Sigma_\theta = \frac{N}{\pi\,\theta_{\rm max}^2} \quad [\,{\rm arcmin}^{-2}], \tag{1}$$

where N is the number of member galaxies (or the richness λ) and $\theta_{\rm max}$ is the angular radius corresponding to the maximum projected distance $R_{\rm max}$ among the members.

3.2 Physical-to-angular conversion

The member distances R are given in $h^{-1}\,{\rm Mpc}$ units. We convert to physical Mpc using $R_{\rm phys} = R/h$ with $h = 0.7$, and then to an angular separation via:

$$\theta = \frac{R_{\rm phys}}{d_A(z)}, \tag{2}$$

where $d_A(z)$ is the angular diameter distance computed under a flat ΛCDM cosmology with $H_0 = 70\ {\rm km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_m = 0.3$. The result is converted from radians to arcmin.

3.3 Numerator choices: raw count vs. richness

We compute two versions of Equation 1:

1. $N = N_{\rm mem}$, the raw number of members listed in the member catalog for each cluster;
2. $N = \lambda$, the redMaPPer richness, which applies statistical corrections for background contamination and membership probability weighting.

3.4 Radial decomposition: inner vs. outer

To assess whether fiber contention is concentrated in cluster cores, we split each cluster's members at a projected radius of $R_{\rm split} = 150\,h^{-1}\,{\rm kpc}$ ($= 0.15\,h^{-1}\,{\rm Mpc}$):

- **Inner region** ($R \le R_{\rm split}$): $\Sigma_{\rm inner} = N_{\rm inner}\,/\,(\pi\,\theta_{\rm split}^2)$.
- **Outer region** ($R > R_{\rm split}$): $\Sigma_{\rm outer} = N_{\rm outer}\,/\,\left[\pi\,(\theta_{\rm max,outer}^2 - \theta_{\rm split}^2)\right]$, where $\theta_{\rm max,outer}$ is the angular radius corresponding to the maximum R among outer members.

Here, $\theta_{\rm split}$ is the angular equivalent of $R_{\rm split}$ at the cluster redshift (Equation 2).

4 Results

4.1 Total angular density

Table 1 summarizes the median angular densities across all 23,801 clusters. The raw-count density ($N_{\rm mem}$-based) exceeds the richness-based density because the member catalog includes galaxies that survive the initial selection but carry membership probability < 1 and would be down-weighted in λ.

Table 1: Median member galaxy angular densities for redMaPPer SDSS DR8 clusters at $z_\lambda \leq 0.5$.

Density estimator	Median [arcmin^{-2}]
Total ($N_{\rm mem}$)	1.14
Total (λ)	0.71
Inner ($R \leq 150\,h^{-1}$ kpc)	4.19
Outer ($R > 150\,h^{-1}$ kpc)	0.99
DESI fiber density	0.20
Stage-V fiber density	1.11

4.2 Redshift dependence

Figure 1 shows the angular density as a function of cluster redshift. Both the $N_{\rm mem}$-based and λ-based densities rise steeply from $z \sim 0.05$ to $z \sim 0.2$–0.3 and flatten at higher redshift. This behavior is driven by the shrinking angular diameter distance: at higher redshift, the same physical extent subtends a smaller angle, compressing members into a smaller angular area.

Nearly all clusters lie above the DESI fiber density threshold at all redshifts. The Stage-V threshold intersects the bulk of the distribution, with most clusters above it for $z \gtrsim 0.15$ in the $N_{\rm mem}$ case and $z \gtrsim 0.25$ in the λ case.

4.3 Inner vs. outer regions

Figure 2 compares the inner and outer angular densities. The inner $150\,h^{-1}$ kpc core has a median density of $4.19\,\mathrm{arcmin}^{-2}$, roughly $4\times$ higher than the outer annulus ($0.99\,\mathrm{arcmin}^{-2}$). The inner density exceeds both the DESI and Stage-V fiber densities across essentially the entire redshift range, confirming severe fiber contention in cluster cores even for Stage-V.

The outer region is more favorable: its density falls between the DESI and Stage-V thresholds for many low-redshift clusters ($z \lesssim 0.15$) and rises above the Stage-V line at $z \gtrsim 0.2$–0.3. This indicates that a single-pass Stage-V observation can achieve high spectroscopic completeness for *outer* satellite galaxies at $z \lesssim 0.2$, but multi-pass strategies become necessary at higher redshifts or for core regions.

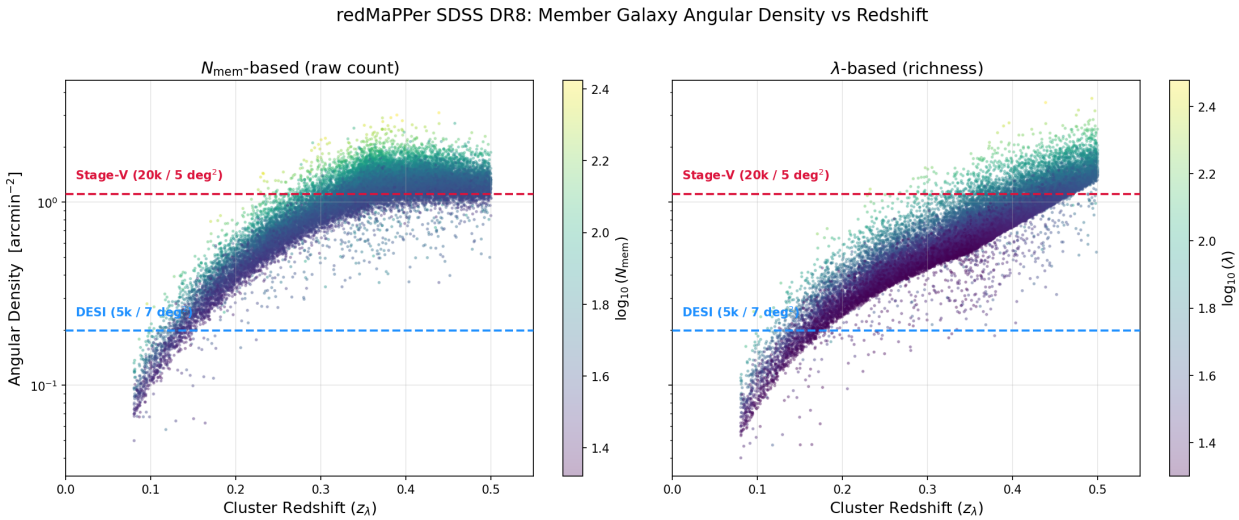

Figure 1: Cluster member galaxy angular density vs. photometric redshift z_λ for 23,801 redMaPPer SDSS DR8 clusters. *Left*: using the raw member count $N_{\rm mem}$. *Right*: using the richness λ. Color encodes the log of the numerator. Horizontal dashed lines mark the DESI (blue; 0.198 arcmin^{-2}) and Stage-V (red; 1.111 arcmin^{-2}) fiber densities.

5 Discussion

These results have several implications for the Spec-S5 survey design:

1. **Core completeness requires multi-pass.** The inner $150\,h^{-1}$ kpc of clusters is $\sim 4\times$ denser than the Stage-V fiber density at all redshifts. Achieving high spectroscopic completeness in cores will require either multiple passes or dedicated cluster-mode observations.

2. **Outskirts are accessible in single-pass at low z.** For $z \lesssim 0.2$, the outer angular density falls near or below the Stage-V fiber density, meaning that satellite spectroscopic redshifts in cluster outskirts can largely be obtained "for free" during the primary survey.

3. **Richness vs. raw counts.** The λ-based density is $\sim 40\%$ lower than the raw-count density, reflecting background subtraction. Since the actual targets for fiber assignment are *candidate* members (not yet background- subtracted), the raw-count density is the more operationally relevant metric for fiber-allocation planning.

4. **Redshift-dependent strategy.** A natural survey strategy would be to rely on single-pass for $z \lesssim 0.2$ clusters and allocate additional passes for higher-redshift systems where the angular density exceeds the fiber density across the full radial extent.

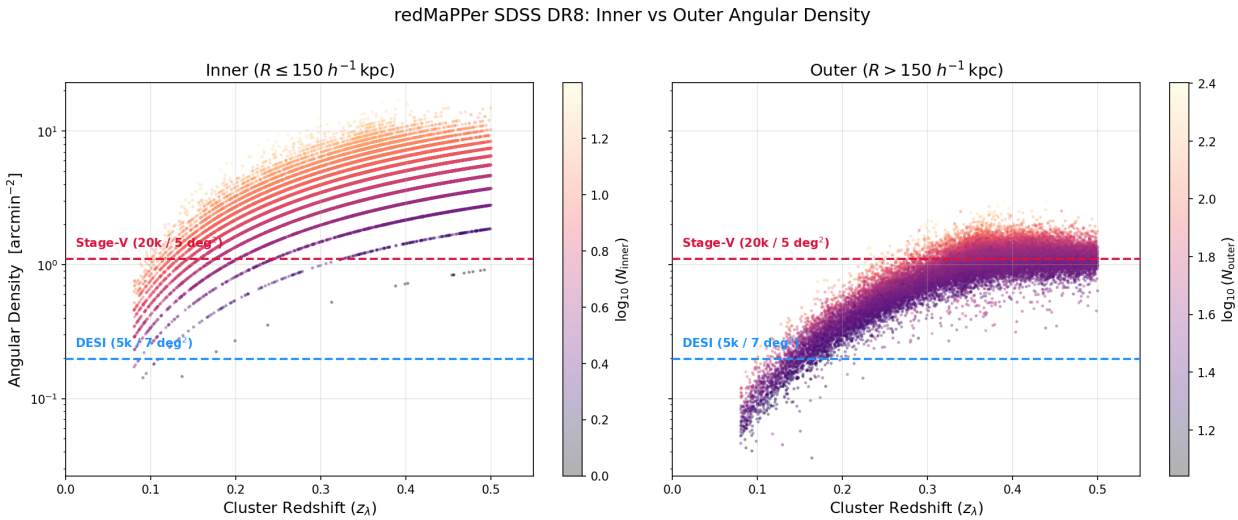

Figure 2: Inner vs. outer angular density for the same cluster sample. *Left*: inner region ($R \leq 150\,h^{-1}$ kpc). *Right*: outer annulus ($R > 150\,h^{-1}$ kpc). Color encodes the log of the member count in each region. Survey fiber density thresholds are shown as in Figure 1.

6 Reproducibility

The analysis is implemented in a single Python script (`angular_density.py`) using `astropy` (FITS I/O and cosmological calculations), `numpy`, and `matplotlib`. The cosmological model is flat ΛCDM with $(H_0,\ \Omega_m) = (70,\ 0.3)$. All per-cluster results (ID, redshift, member counts, angular sizes, and densities for total, inner, and outer regions) are saved to `angular_density_results.dat`.

References

Rykoff, E. S., Rozo, E., Busha, M. T., et al. 2014, ApJ, 785, 104